TEAM BOOGIE

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CS 377Q

P2: Project Proposal

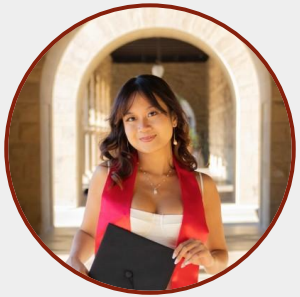
Ari B., Osose E., Mai H., Rachel L., Ananya N.

Hey, we're Team Boogie!

THE TEAM

Ari B.

Osose E.



Mai H.

Rachel L.

Ananya N.

From our research, we found…

NEED-FINDING

The best help comes from **human-enhanced navigation** in forms of voice calls, in-person assistance, conversation, etc.

Sounds that are unique to human communication are more effective cues than more *common sounds produced in an organic environment.*

Most mobile solutions prioritize *turn-by-turn navigation*, while most BLV users would prefer a more **holistic approach** to their environment.

Turn-by-turn navigation **prevents** any form of **exploration**.

Landmarks (entrances, exits, curbs, hazard areas) and clearly defined terrains within an environment help a BLV user in **generating a mental map** of their surroundings, allowing for *more independence in interaction.*

CORE NEED

“How to convert a physical space into a *visualizable* mental representation

Most research currently focuses on creating visual models that have built-out extensions to navigate based on various accessibility needs rather than creating interactable cues that incorporate multi-modality.

New (& improved)!

SOLUTION

Go beyond UI redesign

Instead of focusing on fixing the current UX bugs, explore the opportunity of making a tool specifically design for BLV to get around campus focused on environmental cues.

Scrap the map

We may model campus as a graph, where each location is a node that tied to semantic information, like cardinal directions, proximity to landmarks, and nearby sounds as we figure out pick-up and drop-off locations.

Maximize Other Senses

We will aim to reduce cognitive overload by employing audio-tactile tools such as haptic feedback during navigation and voice agents for gathering user input.

Thank you!

Questions?